Effect Of Blurting Method On Knowledge Acquisition And Recall Performance

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by

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DEDICATION

This study is dedicated to,

The students of Tomas Cabili National High School.

For willingly participating in our study.

Moreover, this study is also encouraged by our advisor, Sir Jonathan R. Paler, Ph.D.

This study would not have been conducted without the support of our families, schoolmates, classmates, and the group of this research.

We dedicate this research to our fathers for supporting us in our journey in this research; without them, we would have been lost in our study.

We also dedicate this research to our mothers for giving us the love we need to continue to do this study and the courage to face challenges.

And lastly, We dedicate this research paper to the almighty God who gives me strength, wisdom, guidance, and power. We thank God for keeping us healthy in this research.

CJCAS, TATA, SAMA, EDS,
RSA, HMB, ADO,
NAE

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CJCAS, TATA, SAMA, EDS,
RSA, HMB, ADO,
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EFFECT OF BLURTING METHOD ON KNOWLEDGE ACQUISITION AND RECALL PERFORMANCE

ABSTRACT

This quantitative experimental study evaluated the effect of the Blurting Method on knowledge acquisition and recall performance among 30 Grade 12 Computer Systems Servicing students. Participants were randomly assigned to an active-retrieval Blurting Group (n = 15) or a conventional study group (n = 15) across a three-day intervention. Assessment data were analyzed using paired and independent samples t-tests. The Blurting Group increased their mean score from 12.93 (SD = 6.58) to 31.00 (SD = 6.00), showing significant gains in knowledge acquisition ($t(14) = -7.47$, $p < .001$) and recall ($t(14) = -6.08$, $p < .001$). The control group shifted from 13.07 (SD = 7.28) to 18.73 (SD = 8.16), showing no significant gain ($p = .058$). Independent t-tests confirmed the Blurting Group significantly outperformed the control group in overall acquisition ($t(28) = 3.39$, $p = .002$) and post-test recall performance ($t(28) = 4.51$, $p < .001$). Active retrieval practice through the Blurting Method substantially improves learning outcomes compared to passive study routines.

Keywords: Keywords: Blurting Method, Active Recall, Knowledge Acquisition, Recall Performance.

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CHAPTER 1

INTRODUCTION

Background of the Study

Learning is an important part of education because students are expected not only to receive information but also to understand and use what they have learned. However, students do not all learn effectively through the same study methods. Some students mainly reread their notes or highlight important words, while others use study methods that require them to actively work with the information. Because of this, finding effective study strategies is important, especially for students who need to learn many technical terms and procedures.

One study strategy that uses active learning is the Blurting Method. The method requires students to study a topic, put away their learning materials, and write down everything they can remember. Afterward, they compare what they wrote with their original material and identify the information they missed or answered incorrectly. This process gives students an opportunity to actively retrieve information instead of simply looking at it repeatedly. The Blurting Method is therefore closely connected to the idea of active recall and retrieval practice.

Retrieval practice is the process of trying to bring previously learned information back from memory. Research by Roediger and Karpicke (2006) showed that actively retrieving information through testing can support learning.

Their findings suggest that testing is not only useful for measuring what students know but can also be part of the learning process. This is important to the present study because Blurting gives students repeated opportunities to retrieve information while studying instead of waiting until the final assessment.

Augustin (2014) also discussed active recall and testing as useful learning approaches. Rather than depending only on repeated reading, students can improve their learning by trying to recall information without immediately looking at the answer. This idea is directly connected to Blurting because the method requires students to hide their notes and produce what they remember on their own.

Xu et al. (2024) reviewed several active recall strategies, including self-testing and retrieval practice, and found that these strategies have potential benefits for academic performance. Their findings support the idea that study methods requiring students to actively retrieve information may help improve their learning. However, their review focused on active recall strategies in general rather than the specific step-by-step process of Blurting.

The idea of desirable difficulty also helps explain why active retrieval may be useful. Bjork and Bjork (2011) explained that some learning activities can feel more difficult but may lead to better learning. Unlike simply rereading notes, Blurting requires students to recall information without looking at their materials. Students may find this difficult, especially when they cannot immediately

remember an answer. However, this effort requires them to actively work with the information instead of relying on familiarity.

This is especially relevant to Computer Systems Servicing (CSS) students. CSS involves learning technical terms, concepts, tools, procedures, and other information that students need to recall during assessments and practical activities. Students may therefore benefit from study methods that require them to produce information from memory rather than simply recognize information while reading.

The Blurting Method may support knowledge acquisition because students are required to actively process and review the information they are unable to recall. After completing the blurt, students compare their answers with their learning material. This allows them to identify gaps in their knowledge and focus their review on information they were unable to produce correctly.

The method may also affect recall performance. Recall refers to the ability to retrieve previously learned information and produce it when needed. This is particularly relevant to identification-type questions because students must provide the correct answer. In the researchers' study, the 28-item identification portion of the test provides a way to examine students' recall performance.

Although active recall and retrieval practice have been studied, there is still limited research focusing specifically on the complete Blurting Method workflow. Many studies examine retrieval practice, testing, flashcards, or other

study strategies separately. There is less research on whether the complete process of studying, hiding the material, writing what is remembered, checking mistakes, and reviewing missed information can improve the performance of Grade 12 students in a technical-vocational setting.

Therefore, this study aims to determine the effect of the Blurting Method on knowledge acquisition and recall performance among Grade 12 Computer Systems Servicing students. The study will compare students who use the Blurting Method with students who use their usual study approach through a pre-test and post-test after a three-day intervention.

Theoretical Framework

This study is anchored on two theories that help explain why the Blurting Method may affect students' learning and recall performance: Retrieval Practice Theory and the Levels of Processing Theory.

Retrieval Practice Theory

Retrieval Practice Theory is associated with the work of Roediger and Karpicke (2006). The theory explains that actively retrieving information from memory can support learning. Instead of repeatedly looking at information, students attempt to produce the information without looking at their study materials.

The Blurting Method directly applies this idea. Students study their material and then hide it before writing down everything they can remember. They are therefore required to retrieve information from memory instead of relying on visible notes.

This theory supports the study because the main activity in the Blurting Method is retrieval. It may help explain why students who use Blurting could show greater improvement in their knowledge acquisition and recall performance than students who do not use the method.

Levels of Processing Theory

The Levels of Processing Theory, proposed by Craik and Lockhart (1972), explains that the way information is processed can affect learning. Information that is processed more actively and meaningfully may be learned better than information that is only processed at a surface level.

The Blurting Method requires students to do more than simply look at their notes. They must recall information, organize what they remember, write it down, compare it with the original material, and correct their mistakes.

This process requires students to actively work with the information. For CSS students, this can be useful because they need to understand and recall technical terms and concepts rather than simply recognize them while reading.

The theory therefore supports the idea that the Blurting Method may help students acquire knowledge and improve their ability to recall learned information during an assessment.

Conceptual Framework

The conceptual framework of this study follows an Input-Process-Output (IPO) model. It shows how the study will examine the effect of the Blurting Method on students' knowledge acquisition and recall performance.

Input

The input includes:

1. **Blurting Group (Experimental Group)**

   Students studied using the Blurting Method for 20 Minutes per day for three consecutive days.

2. **Non-blurting Group (Control Group)**

   Students studied the same learning materials using a non-blurting study approach for the same amount of time.

Before the intervention, both groups will take the same pre-test. The pre-test will establish their starting level of knowledge.

Process

The process involves a three-day study intervention.

The Blurting Group will use the Blurting Method. Students will:

1. Study

Students study the learning material.

2. Hide the Material

Students put away their learning material.

3. Blurt (Write What You Remember)

Students write down everything they can remember.

4. Check

Students compare their blurted notes with the original material.

5. Review

Students review the information they missed or blurted incorrectly.

The Non-blurting Group will study the same material for the same amount of time but will not use the Blurting Method.

Both groups will follow the study schedule for three days. This allows the study to focus on the difference in study method.

Output

After the three-day intervention, both groups will take the post-test.

The study will examine two outcomes:

1. Knowledge Acquisition

This will be examined through the students' overall test scores, particularly the change between their pre-test and post-test performance.

2. Recall Performance

This will be examined through the students' performance on the 28-item identification section, which requires them to produce the appropriate information rather than simply choose an answer.

The results of the two groups will then be compared to determine whether the Blurting Method produces a significant difference in knowledge acquisition and recall performance.

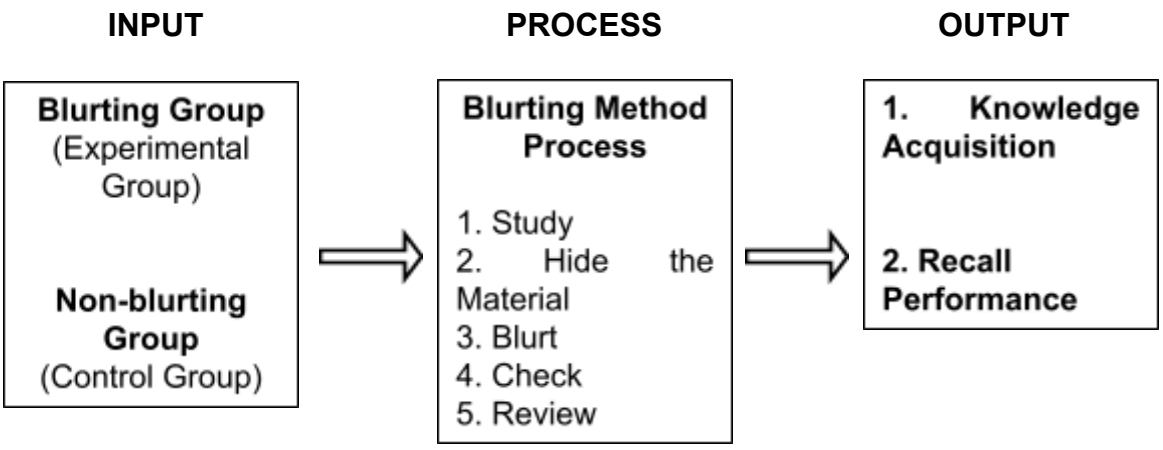


Figure A: The Input-Process-Output model showing how the study will examine the effect of the Blurting Method on students' knowledge acquisition and recall performance.

Statement of the Problem

This study aims to determine the effect of the Blurting Method on knowledge acquisition and recall performance among Grade 12 Computer Systems Servicing students of Tomas Cabili National High School.

Specifically, it seeks to answer the following questions:

1. What is the pre-test performance of the Blurting Group and Non-blurting Group in terms of:

1.1 Overall test score; and

1.2 Identification scores?;

2. What is the post-test performance of the Blurting Group and Non-blurting Group in terms of:

2.1 Overall test score; and

2.2 Identification scores?;

3. Is there a significant difference between the pre-test and post-test results of the Non-blurting Group in terms of:

3.1 Knowledge acquisition; and

3.2 Recall performance?;

4. Is there a significant difference between the pre-test and post-test results of the Blurting Group in terms of:

4.1 Knowledge acquisition; and

4.2 Recall performance?; and

5. Is there a significant difference between the post-test results of the Blurting Group and Non-blurting Group in terms of:

5.1 Knowledge acquisition; and

5.2 Recall performance?

Null Hypothesis

Based on the statement of the problem, the following null hypothesis will be tested:

H_0: There is no significant difference between the post-test results of the Blurting Group and the Non-blurting Group in terms of knowledge acquisition and recall performance.

Significance of the Study

This study is important because it examines whether the Blurting Method can help Grade 12 CSS students improve their learning and ability to recall information compared with a non-blurting study approach.

The findings may benefit the following:

Students. The study may help students understand whether using the Blurting Method can improve their learning and ability to recall information. The method may also help students identify which parts of their lessons they understand and which parts they need to review.

Teachers. The findings may provide teachers with information about the possible usefulness of retrieval-based study activities. If the method produces positive results, teachers may consider introducing Blurting as an additional study strategy for technical lessons.

School Administrators. The findings may provide school administrators with information about an active study strategy that can potentially be used to support students' learning. The results may also help them understand the value of encouraging evidence-based study methods.

Parents and Guardians. The study may help parents and guardians understand that effective studying does not always involve simply reading notes repeatedly. The findings may give them another study strategy that they can encourage students to try at home.

Future Researchers. The study may serve as a reference for future researchers who want to investigate Blurting, retrieval practice, active recall, or other study strategies. Future researchers may also use the study as a starting point for testing the method with different subjects, grade levels, larger groups, or longer intervention periods.

Scope

This study focuses on the effect of the Blurting Method on knowledge acquisition and recall performance among Grade 12 Computer Systems

Servicing students at Tomas Cabili National High School during School Year 2026–2027.

The study will involve 30 Grade 12 CSS students, divided into a Blurting Group and a Non-blurting Group, with 15 students in each group. The study will use a quantitative experimental design and will collect numerical data through pre-test and post-test results. The Intervention will only be done for three-day, 20-minute-per-day study sessions.

The study will focus on selected Computer Networking learning materials. Both groups will receive the same learning materials and the same amount of study time. The main difference between the groups will be the study method used during the intervention.

The Blurting Group will use the complete Blurting Method, while the Non-blurting Group will use a non-blurting study approach.

Knowledge acquisition will be examined using the overall test score, while recall performance will be examined using the 28-item identification section of the assessment.

Delimitation

The study is limited to Grade 12 Computer Systems Servicing students and does not include students from other grade levels or strands. This limitation

allows the researchers to focus on students who are studying technical computer-related concepts.

The study focuses specifically on the Blurting Method and does not compare it with other active study strategies such as flashcards, mind mapping, or spaced repetition.

The study will focus on test-based numerical results and will not include interviews, surveys, or other subjective measures. It will also not examine factors such as intelligence, family income, home environment, or personal motivation.

The study will only examine knowledge acquisition and recall performance using pre-test and post-test scores. It will not measure long-term retention because no delayed post-test will be conducted.

The study will also not examine changes in students' motivation, confidence, study preferences, or anxiety. These factors may affect learning but are outside the main focus of the study.

Definition of Terms

For better understanding of the study, the following terms are defined conceptually and operationally:

Blurting Group:

Conceptually, this refers to the group of participants who receive the experimental treatment.

Operationally, this refers to the Grade 12 CSS students who will be trained and required to use the complete Blurting Method during the three-day intervention.

Blurting Method:

Conceptually, the Blurting Method is a study strategy in which a learner studies a topic, hides the study materials, writes down everything they can remember, and then checks their response against the original material.

Operationally, it refers to the study routine used by the experimental group in which students study the given Computer Networking materials, hide their notes, write what they remember, check their responses, and review missed information.

Computer Systems Servicing (CSS):

Conceptually, Computer Systems Servicing is a technical-vocational field involving the installation, configuration, maintenance, and troubleshooting of computer systems, networks, and related devices.

Operationally, it refers to the Grade 12 CSS students participating in the study and the Computer Networking lessons used as the learning content.

Knowledge Acquisition:

Conceptually, knowledge acquisition refers to the process of gaining new information and understanding through learning.

Operationally, it refers to the students' learning gains as shown by the change in their overall pre-test and post-test scores.

Non-blurting Group:

Conceptually, this refers to the control group that does not receive the experimental treatment.

Operationally, it refers to the Grade 12 CSS students who will study the same learning materials for the same amount of time but will not use the Blurting Method.

Post-test:

Conceptually, a post-test is an assessment given after an intervention to determine students' performance after the learning activity.

Operationally, it refers to the 35-item assessment administered to both groups after the three-day intervention.

Pre-test:

Conceptually, a pre-test is an assessment given before an intervention to determine students' initial level of knowledge.

Operationally, it refers to the 35-item assessment administered before the three-day study intervention.

Recall Performance:

Conceptually, recall performance refers to a learner's ability to retrieve and produce previously learned information.

Operationally, it refers to the students' scores on the 28-item identification section of the pre-test and post-test, where students must provide the appropriate answer without relying on multiple-choice options.

Retrieval Practice:

Conceptually, retrieval practice refers to the process of actively bringing previously learned information back from memory.

Operationally, it refers to the recall activity used in the Blurting Method, particularly when students hide their study materials and write down the information they can remember.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Review of Related Literature

This section presents the concepts and ideas that explain how the Blurting Method may help students learn information and recall it during an assessment. The discussion focuses on active learning, retrieval practice, knowledge acquisition, and recall performance.

Active Learning and Knowledge Acquisition

Learning is not only about reading or looking at information repeatedly. Students learn better when they actively work with the information they are studying. Active learning requires students to participate in the learning process by thinking about, explaining, practicing, or retrieving information instead of only reading it.

Knowledge acquisition refers to the process of gaining new information, concepts, and understanding through learning. In this study, knowledge acquisition refers to the knowledge gained by the students after using the Blurting Method, as shown by the difference between their pre-test and post-test performance. Augustin (2014) explained that effective learning methods include testing oneself and actively recalling information, Improve student's learning by moving away from simply reading their materials and instead trying to retrieve

information from memory. This idea is important to the researchers' study because the Blurting Method requires students to study a topic and then put their materials away. They must then write down what they remember without looking at their notes. This makes the students actively participate in their own learning instead of simply reading the material repeatedly.

A systematic review by Xu et al. (2024) also examined active recall strategies such as flashcards, practice testing, retrieval practice, and concept mapping showed potential for improving academic performance, while self-testing and retrieval practice were useful but not always commonly used by students. This supports the idea that study methods involving active recall may help students gain knowledge more effectively. However, the review focused on different active recall strategies rather than the complete Blurting Method. Therefore, it does not directly show whether Blurting itself improves knowledge acquisition among Grade 12 Computer Systems Servicing students.

Retrieval Practice and Learning

Retrieval practice refers to the act of trying to bring previously learned information back from memory. Instead of looking at the answer, students try to remember the information on their own.

Roediger and Karpicke (2006) explained that testing can do more than measure what students know. Their experiments showed that retrieving information through testing can improve learning. Students who practiced

retrieving information showed stronger performance on later tests than students who only studied the material repeatedly, particularly when the test was given after a delay. Although their research also examined later retention, the important point for the researchers' study is that retrieving information is itself a learning activity. This is closely related to Blurting because students repeatedly attempt to retrieve information without looking at their study materials.

Roediger and Karpicke (2006) also described the testing effect, which refers to the benefit that comes from practicing retrieval. Their findings suggest that a test can serve not only as an assessment but also as part of the learning process. In the researchers' study, Blurting can be viewed as a form of informal retrieval practice. Students do not wait until the final examination to retrieve information. Instead, they practice retrieving information during their study sessions. This may help explain why students using Blurting could show greater improvement between their pre-test and post-test scores.

Active Recall and Recall Performance

Recall is the ability to retrieve information from memory when it is needed. Recall performance refers to how successfully a student can produce previously learned information. This is different from simply recognizing an answer. For example, when a student is given several choices and selects the correct one, the student may recognize the answer. However, when the student is given a description and has to produce the correct term without answer choices, the

student must retrieve the information. This distinction is important to the researchers' study because the questionnaire contains an identification section in which students are required to provide the correct technical terms. Therefore, the identification portion can be used to examine recall performance.

Augustin (2014) emphasized active recall as an effective learning approach. Instead of repeatedly reading information, students are encouraged to test themselves and attempt to remember the information without immediately looking at the answer. The Blurting Method follows this idea. Students first study the material and then hide their notes. They write down as much information as they can remember. Afterward, they compare their answers with the original material and identify the information they missed.

The process gives students an opportunity to practice retrieving information before taking the actual post-test. Therefore, it is reasonable to examine whether students who use Blurting show better recall performance on the identification portion of the assessment.

The Testing Effect and Recall

The testing effect is closely connected to recall because it shows that attempting to retrieve information can affect later performance. Roediger and Karpicke (2006) found that students who practiced retrieval benefited from the act of testing itself. This supports the use of recall-based activities in learning. When students try to remember information without looking at their notes, they

are practicing the same basic process that is required when answering questions that require them to produce an answer.

This is particularly relevant to the 28-item identification section. The students must provide technical terms based on the descriptions given in the test. Because they cannot simply choose an answer from a list, the task requires them to retrieve the information they learned.

Therefore, the identification score can provide an indication of students' recall performance at the time of the post-test.

Desirable Difficulty and Active Study

Bjork and Bjork (2011) discussed the idea of desirable difficulties, which means that some learning activities may feel more difficult while actually helping learning. Easy study activities can sometimes make students feel that they understand the material simply because the information looks familiar. For example, rereading the same notes several times may feel easy because the student becomes familiar with the words. However, familiarity does not always mean that the student can produce the information without assistance.

Blurting introduces difficulty because students have to study without relying on their notes when they try to recall information. They may struggle to remember some concepts, but this struggle is part of the learning activity.

The checking stage of Blurting is also important. After attempting to recall the information, students compare their responses with their study material. This allows them to identify what they knew, what they missed, and what they need to study again. This process may support knowledge acquisition because students are not only receiving information but are also checking their own understanding.

Blurting Method as a Retrieval-Based Study Strategy

The Blurting Method combines several actions into one study routine. First, students study the material. Second, they put away their notes. Third, they write down everything they can remember. Finally, they compare their answers with the original material and review the information they missed. This process is closely related to retrieval practice because students must recall information without looking at their materials.

The method also gives students immediate information about what they know and do not know. Instead of assuming that they understand the material because it looks familiar, they can see which parts they can actually produce without assistance. Because of this, Blurting may affect both knowledge acquisition and recall performance. The overall test score can show whether students improved after the intervention, while the identification section can provide a more specific measure of their ability to recall technical information.

Review of Related Studies

This section presents studies and research reviews related to retrieval practice, active recall, and study strategies. These studies provide evidence that supports the possible use of Blurting as a learning strategy.

Active Recall Strategies and Academic Performance

Xu et al. (2024) conducted a systematic review of active recall strategies and their relationship with academic achievement among young adults. The researchers reviewed studies involving strategies such as flashcards, practice testing, retrieval practice, and concept mapping. Their findings showed that active recall strategies have potential to support academic performance. Self-testing and retrieval practice were also identified as useful strategies, although they were not always commonly used by students.

This study is relevant to the researchers' research because Blurting also requires students to actively retrieve information. However, Xu et al. (2024) examined a variety of active recall strategies rather than the specific Blurting Method.

This creates a gap that the study seeks to address. Instead of examining active recall in general, the researchers' study focuses on a specific written retrieval routine and tests its effect among Grade 12 Computer Systems Servicing students.

Testing as a Learning Activity

Roediger and Karpicke (2006) conducted experiments to compare repeated testing with repeated studying. Students studied passages and either practiced retrieving the information or studied the material again. Their findings showed that repeated retrieval could produce better performance on later tests than repeated studying, especially when the final assessment was delayed. The study supports the idea that retrieval is not only useful for checking knowledge. It can also contribute to learning.

This is related to Blurting because students repeatedly attempt to retrieve information before checking their notes. However, the Roediger and Karpicke study focused on testing and later retention, while the researchers' study focuses on knowledge acquisition and recall performance immediately after a three-day intervention.

Retrieval Practice in the Classroom

A classroom experiment by researchers examining the self-regulated use of retrieval practice investigated whether students could be encouraged to use retrieval practice as a study strategy. The study was conducted in an actual higher-education classroom rather than only in a laboratory setting. The researchers found that strategy instructions could partly support students' use of retrieval practice, although the effect on retrieval-practice use was small and no relationship with performance was found in that particular study.

This study is useful because it shows that retrieval practice can be studied in a real classroom environment. It also shows that simply telling students about a study strategy does not always guarantee improved performance.

This is different from the researchers' study because the Blurting Method will be directly implemented as an intervention. The students in the experimental group will actually use the strategy during the three-day intervention.

Retrieval Practice and New Learning

Research on retrieval-based learning has found that practicing retrieval can support new learning. A review of retrieval practice research explains that attempting to recall previously studied information can produce more learning than simply studying the material again.

This is important to the researchers' study because Blurting requires students to attempt recall even when they are not completely sure of the answer. Afterward, they can check their responses and correct their mistakes.

The process may therefore help students identify areas where their knowledge is incomplete and provide another opportunity to learn the correct information.

Active Recall and Learning Strategies

Augustin (2014) reviewed learning methods that can help students study factual information. The article discussed the testing effect, active recall, and

spaced repetition as useful learning approaches. It emphasized that students should actively test themselves instead of relying only on passive study methods.

This study is related to the researchers' research because Blurting is based on the same basic idea of active recall. However, the researchers' research is different because it applies a specific three-day Blurting intervention to Grade 12 Computer Systems Servicing students and measures their performance using pre-test and post-test results.

Synthesis of the Related Literature and Studies

The reviewed literature and studies show that active retrieval is an important part of effective learning. Rather than only reading and reviewing information, students can benefit from trying to retrieve information from memory.

Augustin (2014) discussed active recall and testing as useful learning methods, while Roediger and Karpicke (2006) provided experimental evidence showing that retrieval can contribute to learning. Xu et al. (2024) further showed that active recall strategies such as self-testing and retrieval practice have poential benefits for academic performance.

The idea of desirable difficulties also helps explain why Blurting may be useful. The method makes studying more effortful because students cannot depend on their notes while recalling information. This difficulty may encourage students to actively work with the information rather than simply becoming familiar with it. Taken together, these sources support the basic idea behind the

Blurting Method: students learn by actively trying to retrieve what they have studied.

However, there is still a gap in the existing literature. Most of the studies and reviews focus on retrieval practice, testing, flashcards, or active recall in general. They do not specifically examine the complete Blurting process of studying, hiding the material, writing what is remembered, checking the answer, and reviewing missed information. This also identifies the lack of localized experimental research testing this complete workflow.

There is also a need to examine this strategy among students in a technical-vocational setting, particularly Grade 12 Computer Systems Servicing students. Students in this setting need to learn technical terms, concepts, and procedures. The current study therefore provides an opportunity to examine whether a retrieval-based study method can improve their learning.

The researchers' study addresses this gap by examining the effect of the Blurting Method on knowledge acquisition and recall performance among Grade 12 Computer Systems Servicing students. Knowledge acquisition will be examined through the overall test performance, while recall performance will be examined through the identification portion of the test.

CHAPTER 3

METHODOLOGY

Research Design

This study used a Quantitative Experimental Research Design. This design allowed the researchers to collect numerical data and determine whether there was a significant difference in the students' performance before and after the intervention.

The study involved two groups; the Blurting Group and the Non-blurting Group. Both groups took the same pre-test before the intervention and the same post-test after the three-day study period. The students were randomly assigned to the two groups using the fishbowl technique. Random assignment gave each selected student an equal chance of being placed in either group. This helped reduce the possible effect of differences in students' background knowledge, learning ability, and study habits.

The main difference between the two groups was the study method used during the intervention. The Blurting Group used the Blurting Method, while the Non-blurting Group used their usual non-blurting study approach.

The researchers then compared the pre-test and post-test results to determine changes in knowledge acquisition and recall performance.

Research Setting

The study was conducted at Tomas Cabili National High School (TCNHS) in Iligan City, Lanao del Norte. The school was established in 1971 and offered education to students from different grade levels and strands.

The study specifically focused on Grade 12 TVL 3 – Computer Systems Servicing (CSS) students enrolled during School Year 2026–2027. The students were randomly assigned to two groups: the Blurting Group and the Non-blurting Group.

The school was selected as the research setting because it had Grade 12 CSS students whose lessons included technical concepts related to computer systems and networking.

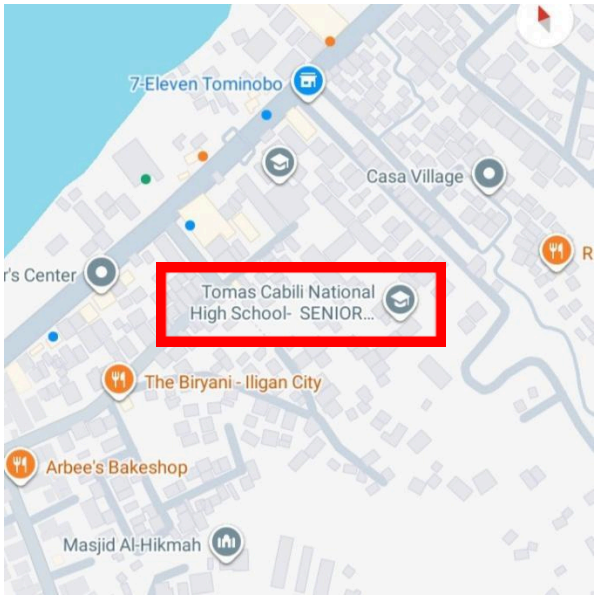


Figure B: The Locale of the Study

Population and Sample

The population of the study consisted of Grade 12 Senior High School students under the Technical-Vocational-Livelihood (TVL) Track at Tomas Cabili National High School during School Year 2026–2027.

The study specifically focused on the Grade 12 TVL 3 – Computer Systems Servicing (CSS) class. The class consisted of 33 students. From this group, 30 students were selected as participants for the study.

The 30 selected participants were divided equally into two groups:

- 15 students – Blurting Group
- 15 students – Non-blurting Group

The sample size allowed the researchers to maintain two balanced groups during the experimental intervention.

Sampling Procedure

The researchers used stratified random sampling to select and divide the participants.

There were 33 students in the Grade 12 TVL 3 – CSS class. Thirty students participated in the study, while three students were not included.

The fishbowl technique was used for the random assignment. The researchers prepared 15 pieces of paper labeled “EXP”, 15 pieces labeled “CON”, and 3 pieces labeled “NONE.” The papers were folded and mixed in a

container. Each student randomly picked one paper. Students who received “EXP” were assigned to the Blurting Group, while students who received “CON” were assigned to the Non-blurting Group. The three students who received “NONE” were not included in the experiment.

The final groups consisted of 15 students each. The Blurting Group was trained to follow the complete Blurting Method. The Non-blurting Group used their usual non-blurting study approach.

Research Instrument

The researchers used an adapted standardized test as the research instrument. The test was administered as both the pre-test and post-test. The test focused on selected Computer Networking topics appropriate to the Grade 12 CSS students. The same test was used before and after the intervention to determine changes in students’ performance.

The learning material used in the study was adapted from the GeeksforGeeks article “Types of Computer Networks” (GeeksforGeeks, 2026). The researchers used the information presented in the material as the basis for developing the 35-item assessment.

The instrument contained 35 items divided into two parts:

Part I – Multiple Choice, consisted of 7 multiple-choice questions. Each question had four choices: A, B, C, and D.

Part II – Identification, consisted of 28 identification questions. Students were required to provide specific technical terms based on the descriptions given.

The 28-item identification section was used to measure recall performance because students had to produce the appropriate information without selecting from multiple-choice answers. The overall 35-item score was used to measure knowledge acquisition, based on the students' change in performance from the pre-test to the post-test. Each correct answer was given one point, giving a maximum possible score of 35 points.

Research Intervention

The Blurting Method served as the experimental intervention. The purpose of the intervention was to provide students with a study method that required them to actively retrieve information instead of relying only on passive study activities such as rereading or highlighting.

The Blurting Group followed four main steps:

Step 1: Study

Students reviewed a specific section of the Computer Networking learning material for a short period.

Step 2: Recall and Write

Students hid their notes and other study materials. They then wrote down everything they could remember about the topic on a blank sheet of paper.

Step 3: Check

Students opened their learning materials and compared what they wrote with the original information. They used a different-colored pen to identify missing or incorrect information.

Step 4: Review Missed Information

Students reviewed the information that they were unable to recall or had answered incorrectly. This allowed them to focus their study on areas where their knowledge was incomplete.

These steps were repeated during the intervention period. Both groups studied for 20 minutes per day for three consecutive days. The sessions were conducted at the same time and in the same classroom setting. Both groups received the same learning materials and the same amount of study time.

The only major difference between the groups was the study method. The Blurting Group used the Blurting Method, while the Non-blurting Group used their usual non-blurting study approach. This arrangement allowed the researchers to determine whether differences in the groups' results were associated with the study method.

Data Gathering Procedure

The researchers followed the following procedures in gathering the data:

1. Request for Permission

The researchers submitted a formal letter to the principal of Tomas Cabili National High School requesting permission to conduct the study.

2. Coordination and Orientation

After approval, the researchers coordinated with the CSS adviser. The researchers conducted an orientation for the students to explain the purpose, procedures, and general mechanics of the study. The researchers also explained that the information collected would be treated confidentially.

3. Random Assignment

The fishbowl technique was used to randomly assign the selected students to the Blurting Group and Non-blurting Group.

4. Pre-test

All 30 participants took the 35-item pre-test before the intervention. The pre-test established the students' initial performance in the selected Computer Networking topics.

The researchers recorded both:

- The overall 35-item score, and
- The 28-item identification score.

5. Intervention

The researchers provided both groups with the same Computer Networking learning materials; The Blurting Group was trained to use the Blurting Method and the Non-blurting Group used their usual non-blurting study approach. Both groups studied for 20 minutes each day for three days under the supervision of the researchers.

6. Post-test

After the three-day intervention, both groups took the 35-item post-test. The researchers scored each correct answer as one point.

7. Organization of Data

The researchers recorded and organized the scores in a spreadsheet.

Two main measures were obtained:

Knowledge Acquisition:

The change in the overall 35-item score from the pre-test to the post-test.

Recall Performance:

The performance on the 28-item identification section of the pre-test and post-test.

Statistical Treatment of Data

The collected data were tabulated, analyzed, and interpreted using appropriate statistical tools. The researchers used a 0.05 level of significance in testing the hypotheses.

1. Mean and Standard Deviation. The mean was used to determine the average pre-test and post-test scores of the Blurting Group and Non-blurting Group. The standard deviation was used to determine how spread out the students' scores were around the mean. These statistics described the groups' performance in terms of:

Overall test scores for knowledge acquisition; and Identification scores for recall performance. These statistical measures addressed Statement of the Problem Nos. 1 and 2.

2. Paired Samples t-test. A paired samples t-test was used to compare the pre-test and post-test scores within each group. It determined whether there was a significant change in:

Knowledge acquisition; and Recall performance. From before to after the three-day intervention. This addressed Statement of the Problem Nos. 3 and 4.

3. Independent Samples t-test. An independent samples t-test was used to compare the post-test results of the Blurting Group and Non-blurting Group. It determined whether there was a significant difference between the two groups in:

Knowledge acquisition, based on overall test performance; and Recall performance, based on the 28-item identification score. This addressed Statement of the Problem No. 5.

CHAPTER 4

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents, analyzes, and interprets the data gathered from the 30 Grade 12 Computer Systems Servicing (CSS) students who participated in the study. The participants were divided into two groups: the Blurting Group and the Non-blurting Group, with 15 students in each group.

The data were gathered through a 35-item pre-test and post-test. The test consisted of 7 multiple-choice items and 28 identification items. The overall test scores were used to examine knowledge acquisition, while the scores from the 28-item identification section were used to examine recall performance. The study followed a three-day intervention, during which the Blurting Group used the Blurting Method while the Non-blurting Group used a non-blurting study approach. Both groups studied the same learning materials for the same amount of time. After the intervention, both groups took the post-test.

The data were presented according to the statement of the problems. Descriptive statistics, including the mean and standard deviation, were used to describe the students' performance. Paired samples t-tests were used to determine whether there were significant changes between the pre-test and post-test scores within each group. An independent samples t-test was used to determine whether there were significant differences between the two groups.

The findings were interpreted in relation to the Retrieval Practice Theory, Testing Effect, Levels of Processing Theory, and related studies discussed in Chapter 2. The results focused on students' knowledge acquisition and recall performance immediately after the three-day intervention.

Statement of the Problem Number 1: What are the pre-test results of the respondents in the Blurting and Non-blurting Groups in terms of:

Table 1.1

Overall Test Scores as the Baseline Measure of Knowledge Acquisition

Group	N	Mean	SD	Maximum Score
Non-blurting Group	15	13.07	7.46	35
Blurting Group	15	12.93	6.65	35

Interpretation

The Non-blurting Group obtained a pre-test mean score of 13.07 (SD = 7.46), while the Blurting Group obtained a mean score of 12.93 (SD = 6.65) out of 35. The difference between the two groups was only 0.14 point, showing that their initial overall test performances were very similar before the intervention.

This similar starting point was important because it provided a reasonable basis for comparing the groups after the three-day intervention. Since both

groups started with nearly the same mean score, the difference in their post-test performance could be examined in relation to the study methods used during the intervention.

This finding also supports the experimental design of the study. The pre-test served as the baseline measure of the students' knowledge before they were exposed to their respective study methods. Therefore, the results provided a starting point for examining changes in knowledge acquisition after the intervention.

Table 1.2

Identification Scores as the Baseline Measure of Recall Performance

Group	N	Mean	SD	Maximum Score
Non-blurting Group	15	10.87	6.38	28
Blurting Group	15	10.67	6.63	28

Interpretation

For the 28-item identification section, the Non-blurting Group obtained a pre-test mean score of 10.87 (SD = 6.38), while the Blurting Group obtained a mean score of 10.67 (SD = 6.63) out of 28. The difference between the groups was only 0.20 point.

The small difference showed that the two groups had very similar initial performance in answering questions that required them to produce specific technical terms. This was important because the identification section was used as the measure of recall performance.

The results therefore provided a similar starting point for both groups before the intervention. Any changes in identification performance observed after the three-day intervention could then be examined in relation to the study method used by each group.

Statement of the Problem Number 2: What are the post-test results of the respondents in the Blurting and Non-blurting Groups in terms of:

Table 2.1

Overall Test Scores as a Measure of Knowledge Acquisition

Group	N	Mean	SD	Maximum Score
Non-blurting Group	15	18.73	8.11	35
Blurting Group	15	31.00	6.07	35

Interpretation

After the three-day intervention, the Non-blurting Group obtained a post-test mean score of 18.73 (SD = 8.11), while the Blurting Group obtained a much higher mean score of 31.00 (SD = 6.07) out of 35.

The Blurting Group therefore obtained an average score 12.27 points higher than the Non-blurting Group.

The higher post-test performance of the Blurting Group may be explained by the active retrieval involved in the Blurting Method. During the intervention, students were required to study the material, hide their notes, write what they remembered, check their answers, and review information they missed. This gave them repeated opportunities to actively work with the information instead of simply reading it.

This finding is consistent with Retrieval Practice Theory, which explains that actively retrieving previously studied information can support learning. Roediger and Karpicke (2006) found that retrieval activities can improve later test performance because students practice bringing information back from memory.

The result is also consistent with Levels of Processing Theory. The Blurting Group did not simply look at the material. They had to recall, organize, write, check, and correct information. These activities required students to actively process the material, which may have contributed to their higher overall post-test performance.

Thus, the results suggest that the Blurting Method was associated with greater knowledge acquisition than the non-blurting study approach used by the other group.

Table 2.2

Identification Scores as a Measure of Recall Performance

Group	N	Mean	SD	Maximum Score
Non-blurting Group	15	15.33	6.59	28
Blurting Group	15	25.00	5.04	28

Interpretation

For the 28-item identification section, the Non-blurting Group obtained a post-test mean score of 15.33 (SD = 6.59), while the Blurting Group obtained a mean score of 25.00 (SD = 5.04).

The Blurting Group scored an average of 9.67 points higher than the Non-blurting Group.

The higher score of the Blurting Group may be related to the fact that the Blurting Method gave students repeated practice in retrieving information without looking at their notes. This was similar to what they were required to do in the identification portion of the post-test, where they had to produce the appropriate technical terms without multiple-choice options.

This finding is consistent with the Testing Effect discussed in Chapter 2. Retrieval-based activities can help students practice producing information rather than simply recognizing it. The identification portion of the assessment required this same type of response.

Therefore, the results suggest that the Blurting Method was associated with better recall performance on the post-test identification section.

Statement of the Problem Number 3: Is there a significant difference between the pre-test and post-test results of the Non-blurting Group in terms of:

Table 3.1

Knowledge Acquisition of the Non-blurting Group

Measure	Pre-test Mean	Post-test Mean	Mean Change	t	df	p	Decision
Overall test score	13.07	18.73	5.67	-2.07	14	.058	Not Significant

Interpretation

The Non-blurting Group's overall mean score increased from 13.07 in the pre-test to 18.73 in the post-test. This represented a mean increase of 5.67 points.

However, the paired samples t-test showed that this increase was not statistically significant, with $t(14) = -2.07$, $p = .058$.

Since the p-value of .058 was greater than the .05 level of significance, the null hypothesis was not rejected. This means that although the Non-blurting Group's average score increased after the three-day study period, the increase was not large enough to be considered statistically significant based on the statistical test used.

The finding may suggest that simply studying the material using the non-blurting approach produced some improvement in students' overall scores, but the improvement was not strong enough to establish a significant change in knowledge acquisition.

This result can be related to the ideas discussed by Bjork and Bjork (2011), who explained that some study activities may feel easier because the material becomes familiar, but familiarity does not always mean that students can actively produce the information when needed. The finding also relates to Levels of Processing Theory because study activities that involve less active processing may provide fewer opportunities for students to work with information deeply.

Table 3.2:

Recall Performance of the Non-blurting Group

Measure	Pre-test Mean	Post-test Mean	Mean Change	t	df	p	Decision
Identification score	10.87	15.33	4.47	-1.88	14	.082	Not Significant

Interpretation

The Non-blurting Group's mean identification score increased from 10.87 in the pre-test to 15.33 in the post-test. This represented a mean increase of 4.47 points.

However, the paired samples t-test showed that the increase was not statistically significant, with $t(14) = -1.88$, $p = .082$.

Since the p-value was greater than .05, the null hypothesis was not rejected. This means that although the Non-blurting Group showed an improvement in its average identification score, the change was not statistically significant.

The result suggests that the non-blurting study approach used by the group did not produce a statistically significant improvement in their ability to recall the technical terms included in the identification section.

This finding is relevant to Retrieval Practice Theory because the Non-blurting Group did not receive the same repeated retrieval practice as the Blurting Group. Their study method provided fewer opportunities to practice producing information without looking at their materials.

Statement of the Problem Number 4: Is there a significant difference between the pre-test and post-test results of the Blurting Group in terms of:

Table 4.1

Knowledge Acquisition of the Blurting Group

Measure	Pre-test Mean	Post-test Mean	Mean Change	t	df	p	Decision
Overall test score	12.93	31.00	18.07	-7.47	14	< .001	Significant

Interpretation

The Blurting Group's overall mean score increased from 12.93 in the pre-test to 31.00 in the post-test. This represented a mean increase of 18.07 points.

The paired samples t-test showed that this increase was statistically significant, with $t(14) = -7.47$, $p < .001$.

Because the p-value was less than .05, the null hypothesis was rejected. This indicates that the Blurting Group showed a statistically significant improvement in overall test performance after the three-day intervention.

The large increase may be explained by the active retrieval process used during Blurting. Students were not only exposed to the learning material; they were required to recall information, write it down, compare it with the source material, identify mistakes, and review the information they missed.

This finding is consistent with Retrieval Practice Theory because the students repeatedly practiced retrieving information. It is also consistent with Levels of Processing Theory, since the students actively worked with and organized the information rather than simply reading it. The finding is also related to the concept of desirable difficulty discussed by Bjork and Bjork (2011). Although recalling information without notes can be more difficult than rereading, the effort involved may encourage students to actively engage with the material.

Therefore, the results indicate that the Blurting Group experienced a statistically significant improvement in knowledge acquisition after the intervention.

Table 4.2

Recall Performance of the Blurting Group

Measure	Pre-test Mean	Post-test Mean	Mean Change	t	df	p	Decision
Identification score	10.67	25.00	14.33	-6.08	14	< .001	Significant

Interpretation

The Blurting Group's mean identification score increased from 10.67 in the pre-test to 25.00 in the post-test. This represented a mean increase of 14.33 points.

The paired samples t-test showed that this increase was statistically significant, with $t(14) = -6.08$, $p < .001$.

Because the p-value was less than .05, the null hypothesis was rejected. This result indicates that the Blurting Group demonstrated a statistically significant improvement in recall performance after the three-day intervention.

The improvement may be explained by the fact that Blurting required students to repeatedly retrieve information without looking at their notes. This practice was closely related to the demands of the identification section of the post-test, where students were required to produce specific technical terms.

The result supports the Testing Effect and Retrieval Practice Theory discussed in Chapter 2. Both explain why actively retrieving information can

support learning and later performance. The result is also consistent with Xu et al. (2024), whose review found that active recall strategies such as self-testing and retrieval practice have potential benefits for academic performance.

Statement of the Problem Number 5: Is there a significant difference between the Blurting and Non-blurting Groups in terms of

Table 5.1

Comparison of Knowledge Acquisition Between the Two Groups

Group	N	Mean Learning Gain	SD	t	df	p	Decision
Non-blurting Group	15	5.67	10.61				
Blurting Group	15	18.07	9.37	3.39	28	.002	Significant

Interpretation

The Blurting Group obtained an average learning gain of 18.07 points (SD = 9.37), while the Non-blurting Group obtained an average learning gain of only 5.67 points (SD = 10.61).

The independent samples t-test showed a statistically significant difference between the two groups, $t(28) = 3.39$, $p = .002$.

Since the p-value was less than .05, the null hypothesis was rejected. This means that there was a statistically significant difference between the two groups in terms of knowledge acquisition, with the Blurting Group showing a greater improvement from pre-test to post-test.

The difference may be explained by the active retrieval and checking process involved in the Blurting Method. While students in the Non-blurting Group used their usual study approach, the Blurting Group repeatedly attempted to recall information, checked their answers, and reviewed the information they missed.

This finding supports the literature on retrieval practice discussed in Chapter 2. Roediger and Karpicke (2006) showed that retrieval can contribute to learning, while Xu et al. (2024) reported that active recall strategies have potential benefits for academic performance.

Therefore, the results suggest that the Blurting Method was more effective than the non-blurting study approach in improving knowledge acquisition among the participants of this study.

Table 5.2

Comparison of Recall Performance Between the Two Groups

Group	N	Mean Identification Score	SD	t	df	p	Decision
Non-blurting Group	15	15.33	6.59				
Blurting Group	15	25.00	5.04	4.51	28	< .001	Significant

Interpretation

The Blurting Group obtained a post-test identification mean score of 25.00 (SD = 5.04), while the Non-blurting Group obtained a mean score of 15.33 (SD = 6.59).

The Blurting Group therefore scored an average of 9.67 points higher than the Non-blurting Group.

The independent samples t-test showed a statistically significant difference between the two groups, $t(28) = 4.51$, $p < .001$.

Since the p-value was less than .05, the null hypothesis was rejected. This means that there was a statistically significant difference between the Blurting

Group and Non-blurting Group in terms of recall performance, with the Blurting Group performing better on the identification section.

The finding can be explained by the close match between the Blurting activity and the demands of the identification test. During Blurting, students practiced recalling information without looking at their notes and writing what they remembered. The identification section similarly required students to produce the correct technical terms without multiple-choice options.

This supports the Testing Effect and Retrieval Practice Theory because the Blurting Group repeatedly practiced the same basic process required during the identification assessment: retrieving information and producing an answer.

Therefore, the findings suggest that the Blurting Method was associated with better recall performance than the non-blurting study approach among the participants.

Summary of Findings

The results showed that the Blurting and Non-blurting Groups started with very similar pre-test performance. The overall pre-test means were 12.93 for the Blurting Group and 13.07 for the Non-blurting Group. Their identification pre-test means were also similar, at 10.67 and 10.87, respectively.

After the three-day intervention, the Blurting Group obtained higher post-test means in both measures. The Blurting Group obtained an overall mean

of 31.00, compared with 18.73 for the Non-blurting Group. For the identification section, the Blurting Group obtained a mean of 25.00, compared with 15.33 for the Non-blurting Group.

The Non-blurting Group did not show a statistically significant improvement in overall test performance ($p = .058$) or recall performance ($p = .082$).

In contrast, the Blurting Group showed statistically significant improvements in both knowledge acquisition ($p < .001$) and recall performance ($p < .001$).

The comparison of learning gains also showed a statistically significant difference between the two groups in knowledge acquisition, with the Blurting Group obtaining a mean gain of 18.07 points, compared with 5.67 points for the Non-blurting Group ($p = .002$).

A statistically significant difference was also found between the groups in recall performance, with the Blurting Group obtaining a post-test identification mean of 25.00, compared with 15.33 for the Non-blurting Group ($p < .001$).

Overall, the findings suggest that the Blurting Method was associated with greater knowledge acquisition and better recall performance among the Grade 12 Computer Systems Servicing students who participated in the study. These

findings are consistent with the ideas of Retrieval Practice Theory, the Testing Effect, Levels of Processing Theory, and active recall.

CHAPTER 5

SUMMARY, FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary of our study, the summary of findings from the gathered data, the conclusions we made based on testing the hypotheses, and recommendations for the people who will benefit from this study.

Summary

This study was conducted to determine the effect of the Blurting Method on knowledge acquisition and recall performance among Grade 12 Computer Systems Servicing (CSS) students at Tomas Cabili National High School.

The study used a quantitative experimental research design with a pre-test and post-test. A total of 30 Grade 12 CSS students participated in the study. The participants were divided into two groups: the Blurting Group, consisting of 15 students, and the Non-blurting Group, also consisting of 15 students. Before the intervention, both groups took the same 35-item pre-test consisting of 7 multiple-choice questions and 28 identification questions. The overall test score was used to examine knowledge acquisition, while the 28-item identification section was used to examine recall performance. The Blurting Group underwent a three-day intervention in which the students used the Blurting Method. They studied the learning material, hid their notes, wrote down what they could remember, checked their answers against the original material, and

reviewed the information they missed. The Non-blurting Group studied the same materials for the same amount of time but did not use the Blurting Method. After the three-day intervention, both groups took the same 35-item post-test. The data were analyzed using the mean, standard deviation, paired samples t-test, and independent samples t-test at a 0.05 level of significance.

The findings showed that the Blurting Group had greater improvements in both overall test performance and identification performance compared with the Non-blurting Group.

Findings

Based on the data presented and analyzed in Chapter 4, the following findings were obtained:

1. Pre-test Performance

The two groups had similar performance before the intervention. For the overall 35-item test, the Blurting Group obtained a mean score of 12.93, while the Non-blurting Group obtained a mean score of 13.07. For the 28-item identification section, the Blurting Group obtained a mean score of 10.67, while the Non-blurting Group obtained a mean score of 10.87.

The small differences between the groups indicated that their initial performance was generally similar before the intervention.

2. Post-test Performance

After the three-day intervention, the Blurting Group obtained substantially higher scores than the Non-blurting Group. For the overall test, the Blurting Group obtained a mean of 31.00, while the Non-blurting Group obtained a mean of 18.73. For the identification section, the Blurting Group obtained a mean of 25.00, while the Non-blurting Group obtained a mean of 15.33.

These results showed that the Blurting Group performed better after the intervention in both measures.

3. Knowledge Acquisition of the Non-blurting Group

The Non-blurting Group's overall mean increased from 13.07 in the pre-test to 18.73 in the post-test.

However, the paired samples t-test resulted in $p = .058$, which was greater than the .05 level of significance.

Therefore, the improvement in the Non-blurting Group's overall performance was not statistically significant.

4. Recall Performance of the Non-blurting Group

The Non-blurting Group's identification mean increased from 10.87 in the pre-test to 15.33 in the post-test.

However, the paired samples t-test resulted in $p = .082$, which was greater than .05.

Therefore, the improvement in the Non-blurting Group's recall performance was not statistically significant.

5. Knowledge Acquisition of the Blurting Group

The Blurting Group's overall mean increased from 12.93 in the pre-test to 31.00 in the post-test. The paired samples t-test resulted in $p < .001$, which was less than .05.

Therefore, the Blurting Group demonstrated a statistically significant improvement in knowledge acquisition after the three-day intervention.

6. Recall Performance of the Blurting Group

The Blurting Group's identification mean increased from 10.67 in the pre-test to 25.00 in the post-test. The paired samples t-test resulted in $p < .001$, which was less than .05.

Therefore, the Blurting Group demonstrated a statistically significant improvement in recall performance after the intervention.

7. Difference Between the Two Groups in Knowledge Acquisition

The Blurting Group had a mean gain of 18.07 points, while the Non-blurting Group had a mean gain of 5.67 points. The independent samples t-test resulted in $p = .002$, which was less than .05.

Therefore, there was a statistically significant difference between the two groups in knowledge acquisition. The Blurting Group demonstrated greater improvement than the Non-blurting Group.

8. Difference Between the Two Groups in Recall Performance

The Blurting Group obtained a post-test identification mean of 25.00, while the Non-blurting Group obtained 15.33. The independent samples t-test resulted in $p < .001$, which was less than .05.

Therefore, there was a statistically significant difference between the two groups in recall performance, with the Blurting Group achieving higher performance.

Conclusion

Based on the findings of the study, the following conclusions were drawn:

1. The two groups had similar starting performance.

The pre-test results showed that the Blurting Group and Non-blurting Group had similar overall and identification scores before the intervention. This indicated that the groups had generally comparable starting performance in the selected Computer Networking topics.

2. The Blurting Method was associated with greater improvement in knowledge acquisition.

The Blurting Group showed a statistically significant increase in overall test performance from the pre-test to the post-test. Furthermore, its improvement was significantly greater than that of the Non-blurting Group.

Therefore, the findings suggest that the Blurting Method was effective in improving knowledge acquisition among the participants of the study.

3. The Blurting Method was associated with improved recall performance.

The Blurting Group demonstrated a statistically significant improvement in the 28-item identification section. It also performed significantly better than the Non-blurting Group in the post-test.

Therefore, the findings suggest that the Blurting Method was effective in improving students' recall performance on the assessment used in the study.

4. The results support the use of active retrieval during studying.

The findings suggest that requiring students to actively retrieve information, write what they remember, check their answers, and review missed information may be more beneficial than simply using a non-blurting study approach.

This conclusion is consistent with the principles of retrieval practice and the testing effect, which emphasize the value of actively retrieving information during learning.

Recommendations

Based on the findings and conclusions, the following recommendations were made:

Students. Students may use the Blurting Method as an additional study strategy, especially when studying lessons that require them to recall technical terms, concepts, and important information. They may practice by studying their materials, putting them away, writing everything they remember, checking their answers, and reviewing the information they missed.

Teachers. Teachers may introduce the Blurting Method as an additional learning and study strategy, particularly for subjects that require students to recall definitions, technical terms, concepts, and procedures. Teachers may also combine Blurting with other effective study strategies to provide students with different ways of learning.

School Administrators. School administrators may consider supporting activities that introduce students to active and retrieval-based study strategies. Workshops or study-skills programs may be conducted to help students learn different approaches to studying. This will also serve administrators the

necessary insights to instruct and guide teachers to systematically integrate the Blurting Method into their daily lessons.

Parents and Guardians. Parents and guardians may encourage students to use active study methods at home. They may encourage students to test themselves and recall information instead of relying only on repeated reading of their notes.

Future Researchers. Future researchers may conduct similar studies using a larger sample size, different grade levels, strands, subjects, or schools. They may also compare the Blurting Method with other study strategies such as flashcards, spaced repetition, mind mapping, or other retrieval-based techniques. Future researchers who want to investigate memory retention should include a delayed post-test after a specific period. This would allow them to determine whether the effects of the study method remain after time has passed. They may also consider conducting the intervention for a longer period to determine whether the results remain consistent over time.

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Appendix A

LETTER FOR THE SCHOOL PRINCIPAL

July 28, 2026

CARLITO A. ABARQUEZ, PhD
School Principal
Tomas Cabili National High School
Iligan City

Dear Sir:

We, the Grade 12 CSS student researchers, respectfully request your permission to conduct gathering of data among Grade 12 CSS – TVL 3 students of Tomas Cabili National High School for our experimental research study entitled “Effect of Blurting Method on Knowledge Acquisition and Recall Performance”. The study will be using the pre-test, intervention, and post-test. The post-test will be conducted after three days from the end of the intervention.

We assure you that all information gathered will be confidential and will be used solely for academic purpose. Thank you for taking time in reading our letter, we hope for your approval and support.

Respectfully yours,

Trisha Arabella T. Alivio _____ Raihana S. Acoon _____

Sittie Aina M. Ampuan _____ Eugene D. Subido _____

Hasnia M. Basher _____ Asrima D. Osop _____

Norjana A. Ela _____

Charlyn Jen Cristen A. Sanchez _____

Noted by:

JONATHAN R. PALER
Subject Teacher

Approved by:

CARLITO A. ABARQUEZ, PhD
School Principal

Appendix B

LETTER FOR THE GRADE 12 CSS STUDENTS

Dear Students of the CSS Strand,

We are conducting a research study entitled “Effect of Blurting Method on Knowledge Acquisition and Recall Performance”.

The purpose of this study is to examine whether the Blurting Method is effective in acquiring knowledge and active recall. Your cooperation and participation will help us reach a conclusion whether the Blurting Method is effective or not.

Please be assured of the following:

- All test scores will be treated with strict confidentiality and used solely for academic and research purposes.
- Your Identity will not be disclosed in any report, publication, or presentation resulting from this study.

Kindly sign the consent form below to indicate that you have read and understood the information provided, and that you voluntarily agree to take part in this study.

Thank you for your time and cooperation.

Signature of Correspondents:

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Appendix C

PRE-TEST QUESTIONNAIRE

Computer Network

PRE-TEST

IDENTIFICATION **Group:_____**

DIRECTION: Identify the terms being described below. Write the letter on **the space provided** before the number.

Computer Network	Ownership and Access Control	Metropolitan Area Network
Geographical Area	Personal Area Network	Campus Area Network
Transmission Medium	Local Area Network	Wide Area Network

_____1. A group of two or more independent computers and devices connected to share data, resources, and services.

2-4. 3 Network Classification.

_____2. (Size/scale)

_____3. (how data travels)

_____4. (Who controls and uses it)

5-9. Coverage Range.

_____5. City or large town.

_____6. Room, building, or small campus.

_____7. University or corporate campus.

_____8. Country, continent, or global.

_____9. 1–10 meters.

10-14. Key Technologies.

_____10. Ethernet, Fiber Optics.

_____11. Ethernet, Wi-Fi.

_____12. Bluetooth, NFC, Infrared.

_____13. Leased lines, Satellite links, Internet.

_____14. Fiber Optics, Microwave, Metro Ethernet.

15-19. When It Is Used.

_____15. To enable long-distance and global communication.

_____16. To provide high-speed connectivity across a city.

_____17. To interconnect LANs across a campus under one organization.

_____18. For fast and secure communication within a local location.

_____19. For short-range communication between personal devices.

20-24. Real-World Example.

_____20. Home or office Wi-Fi network.

_____21. The Internet.

_____22. Smartphone connected to wireless earbuds.

_____23. College or corporate campus network.

_____24. City-wide ISP network.

_____25. Categorizing computer networks according to the physical distance or area they cover, ranging from small personal spaces to large global regions.

_____26. Connections can be established using wired media or wireless technologies.

_____27. Focuses on administrative control and accessibility, rather than transmission medium, size, or technology.

_____28. Refers to the method used to send data signals from one device to another.

MULTIPLECHOICE

DIRECTION: Choose the letter of the correct answer. Write your answers on **the space provided** before the number.

____1. It connects the organization's internal network with trusted outsiders.

A. Internetwork **B**. Hub **C**.VPN **D**. Extranet

____2. Uses dedicated links between communicating devices.

A. Extranet **B**. Point-to-Point Network **C**. Cloud Network **D**. Intranet

____3. Common in modern enterprise and cloud-based systems.

A. Private Network **B**. Hybrid Network **C**. Broadcast Network **D**. Extranet

____4. No dedicated physical path exists between sender and receiver.

A. Intranet **B**. Internet **C**. Broadcast Network **D**. Point-to-Point Network

____5. Used mainly for internal communication.

A. Private Network **B**. Internetwork **C**. Cellular Network **D**. Internet

____6. Owned and controlled by a single organization.

A. Hybrid Network **B**. Private Network **C**. Internetwork **D**. Intranet

____7. Offers wide accessibility but lower security.

A. Broadcast Network **B**. Public Network **C**. Extranet **D**. Cloud Network

Appendix D

POST-TEST QUESTIONNAIRE

Computer Network

POST-TEST

IDENTIFICATION **Group:_____**

DIRECTION: Identify the terms being described below. Write the letter on **the space provided** before the number.

Computer Network	Ownership and Access Control	MAN (Metropolitan Area Network)
Geographical Area	PAN (Personal Area Network)	CAN (Campus Area Network)
Transmission Medium	LAN (Local Area Network)	WAN (Wide Area Network)

_____1. Connections can be established using wired media or wireless technologies.

_____2. City-wide ISP network.

_____3. Leased lines, Satellite links, Internet.

_____4. To provide high-speed connectivity across a city.

_____5. Home or office Wi-Fi network.

_____6. (Who controls and uses it)

_____7. Focuses on administrative control and accessibility, rather than transmission medium, size, or technology.

_____8. For short-range communication between personal devices.

_____9. Room, building, or small campus.

_____10. Smartphone connected to wireless earbuds.

_____11. City or large town.

_____12. Bluetooth, NFC, Infrared.

_____13. To interconnect LANs across a campus under one organization.

_____14. (Size/scale)

_____15. Refers to the method used to send data signals from one device to another.

_____16. Ethernet, Fiber Optics.

_____17. The Internet.

_____18. Country, continent, or global.

_____19. A group of two or more independent computers and devices connected to share data, resources, and services.

_____20. Fiber Optics, Microwave, Metro Ethernet.

_____21. Ethernet, Wi-Fi.

_____22. College or corporate campus network.

_____23. 1–10 meters.

_____24. For fast and secure communication within a local location.

_____25. University or corporate campus.

_____26. (how data travels)

_____27. Categorizing computer networks according to the physical distance or area they cover, ranging from small personal spaces to large global regions.

_____28. To enable long-distance and global communication.

MULTIPLECHOICE

DIRECTION: Choose the letter of the correct answer. Write your answers on **the space provided** before the number.

____1. Owned and controlled by a single organization.

A. Hybrid Network **B**. Intranet **C**. Internetwork **D**. Private Network

____2. No dedicated physical path exists between sender and receiver.

A. Broadcast Network **B**. Internet **C**. Intranet **D**. Point-to-Point Network

____3. Uses dedicated links between communicating devices.

A. Extranet **B**. Intranet **C**. Cloud Network **D**. Point-to-Point Network

____4. Offers wide accessibility but lower security.

A. Broadcast Network **B**. Extranet **C**. Public Network **D**. Cloud Network

____5. It connects the organization’s internal network with trusted outsiders.

A. Extranet **B**. Hub **C**. VPN **D**. Internetwork

____6. Common in modern enterprise and cloud-based systems.

A. Private Network **B**. Extranet **C**. Broadcast Network **D**. Hybrid Network

____7. Used mainly for internal communication.

A. Internetwork **B**. Private Network **C**. Cellular Network **D**. Internet

Appendix E

LEARNING MATERIALS

Computer Network. A computer network is a group of two or more independent computers and devices connected to share data, resources, and services. These connections can be established using wired media or wireless technologies. Types of Computer Networks are classified based on several factors:

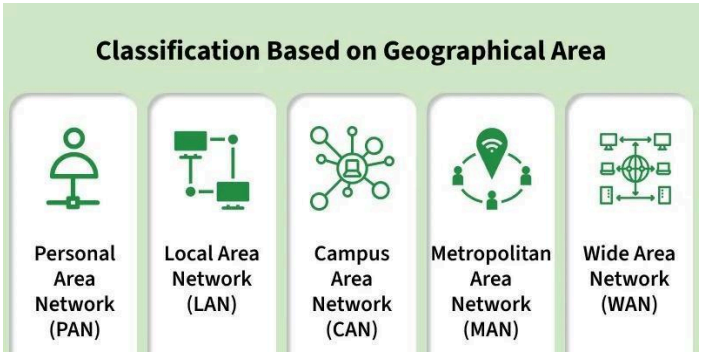
• Geographical area

• Transmission medium

• Ownership and access control

I. Classification Based on Geographical Area

Categorizing computer networks according to the physical distance or area they cover, ranging from small personal spaces to large global regions. Based on geographical area, computer networks are mainly divided into five types:



1. Personal Area Network (PAN)

This connects personal devices within a very short range around a single user.

• Coverage Range: 1–10 meters

• Key Technologies: Bluetooth, NFC, Infrared

• Real-World Example: Smartphone connected to wireless earbuds

• When It Is Used: For short-range communication between personal devices

2. Local Area Network (LAN)

This connects computers and devices within a limited area such as a home, office, or building.

• Coverage Range: One room, building, or small campus

• Key Technologies: Ethernet, Wi-Fi

• Real-World Example: Home or office Wi-Fi network

• When It Is Used: For fast and secure communication within a local location

3. **Campus Area Network (CAN)**
   A CAN connects multiple LANs within a campus or a group of nearby buildings.

• Coverage Range: University or corporate campus

• Key Technologies: Ethernet, Fiber Optics

• Real-World Example: College or corporate campus network

• When It Is Used: To interconnect LANs across a campus under one organization

4. **Metropolitan Area Network (MAN)**
   A MAN connects multiple networks within a city or metropolitan region.

• Coverage Range: City or large town

• Key Technologies: Fiber Optics, Microwave, Metro Ethernet

• Real-World Example: City-wide ISP network

• When It Is Used: To provide high-speed connectivity across a city

5. **Wide Area Network (WAN)**
   A WAN connects networks over large geographical areas such as countries or continents.

• Coverage Range: Country, continent, or global

• Key Technologies: Leased lines, Satellite links, Internet

• Real-World Example: The Internet

• When It Is Used: To enable long-distance and global communication

II. Classification Based on Transmission Medium

Computer networks can also be classified based on how data is transmitted between devices. Transmission technology refers to the method or medium used to send data signals from one device to another.

1. **Broadcast Network**
   This used a shared communication medium among all connected devices.

• Data transmitted by one node is received by all other nodes in the network.

• Each data packet contains a destination address, and only the intended device processes it.

• Communication follows a one-to-many model.

• No dedicated physical path exists between sender and receiver.

Examples:

• Ethernet (bus-based)

• Wireless LAN (Wi-Fi)

2. **Point-to-Point Network**
   This uses dedicated links between communicating devices.

• Data is transmitted directly from one node to another.

• Communication follows a one-to-one model.

• Data may pass through intermediate devices (routers or switches).

• Requires routing algorithms to determine the best path.

Examples:

• Internet

• Leased line networks

• Telephone networks

III. Classification Based on Ownership and Access Control

This classification focuses on administrative control and accessibility, rather than transmission medium, size, or technology. Based on transmission technology, computer networks are mainly divided into three types:

1. **Private Network**
   This type of network is owned and managed by an individual, organization, or institution

• Access is restricted to authorized users only

• Requires authentication such as usernames, passwords, VPNs, or access tokens

• Used mainly for internal communication

• Offers higher security, privacy, and control

Examples:

• Corporate intranet

• Office LAN

• Private cloud network

2. **Public Network**
   This type of network is owned by service providers or government authorities

• Accessible to the general public

• Requires little or no authorization to connect

• Offers wide accessibility but lower security

• Usually shared by a large number of users

• Can operate over both wired and wireless media

Examples:

• Internet

• Public Wi-Fi networks

• Mobile cellular networks

3. **Hybrid Network**
   This type of network is combination of private and public networks

• Some parts of the network are restricted, while others are publicly accessible

• Allows organizations to maintain internal security while providing external connectivity

• Common in modern enterprise and cloud-based systems

Examples:

• Corporate network connected to the Internet

• Enterprise cloud networks

• Online banking systems

Internetwork

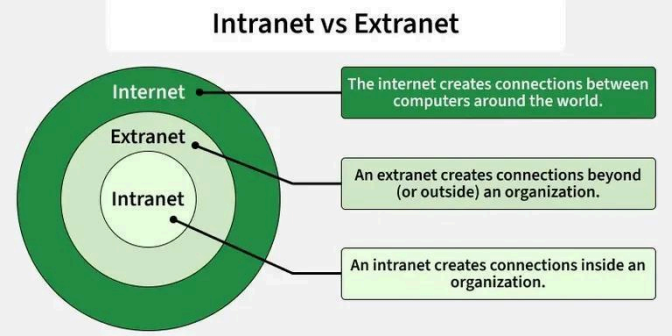
An internetwork is a collection of two or more independent networks connected together to function as a single logical network.

• It enables communication between devices across different networks.

• The Internet itself is the largest example of an internetwork.

• Consists of multiple networks, not just individual computers.

• Uses networking devices to interconnect networks



1. **Intranet**
   An intranet is a private network used within an organization. It is accessible only to authorized users such as employees.

• Owned and controlled by a single organization

• Access restricted using login credentials

• Used for internal communication and collaboration

• Improves information sharing within the organization

Uses:

• Employee portals

• Internal emails and messaging

• Sharing company policies and documents

• Internal applications (HR, payroll, project management)

Examples:

• Corporate internal website

• University campus portal

• Government department internal systems

2. Extranet

An extranet is an extension of an intranet that allows limited access to external users. It connects the organization's internal network with trusted outsiders.

• Partially private network

• Access provided to partners, vendors, suppliers, or customers

• Requires authentication and authorization

• Often implemented using VPN or secure web access

Uses:

• Business-to-business (B2B) communication

• Supply chain management

• Customer portals

• Partner collaboration platforms

Examples:

• Supplier management system

• Online banking portals

• Vendor access portals

Appendix F

DOCUMENTATION



CURRICULUM VITAE



Personal:

Name : Eugene D. Subido

Gender : Male

Civil Status : Single

Address : Purok 16, Tominobo

Place of Birth : Iligan City, Lanao del Norte

Date of Birth : September 30, 2008

Age : 17

Parents : Gemma D. Subido
:

Complexion : Medium

Nationality : Filipino

Religion : Catholic

Height : 4'10”

Weight : 110 lbs.

Academic:

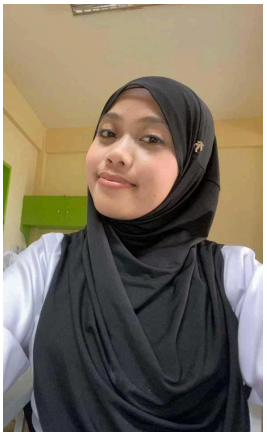
Elementary : Tomas Cabili Central School

High School : Tomas Cabili National High School

Skills:

Gymnastics

CURRICULUM VITAE



Personal:

Name : Raihana S. Acoon

Gender : Female

Civil Status : Single

Address : Mahayahay, Iligan City

Place of Birth : Budong Radapan Matungao, Lanao del Norte

Date of Birth : October 16, 2007

Age : 18

Parents : Acoon Sangcopan
: Casamina Sultan

Complexion : Medium

Nationality : Filipino

Religion : Islam

Height : 4'11”

Weight : 84 lbs.

Academic:

Elementary : Cadayonan School

High School : Tomas Cabili National High School

CURRICULUM VITAE



Personal:

Name : Trisha Arabella T. Alivio
Gender : Female

Civil Status : Single

Address : Purok 1, Sta. Elena, Steel Town, Iligan City

Place of Birth : Iligan City, Lanao del Norte

Date of Birth : May 30, 2009

Age : 17

Parents : Mari Jean T. Alivio
: Joseph B. Alivio

Complexion : Medium

Nationality : Filipino

Religion : Christian

Height : 5’2”

Weight : 97 lbs.

Academic:

Elementary : Tomas Cabili Central School

High School : Tomas Cabili National High School

Skills:

Drawing

Acrylic Painting

CURRICULUM VITAE



Personal:

Name : Sittieaina M. Ampuan

Gender : Female

Civil Status : Single

Address : Tubod, Iligan City

Place of Birth : Tagoranao Bayang, Lanao del Norte

Date of Birth : December 1, 2008

Age : 17

Parents : Norhanifa M. Sarip
: Abdul B. Ampuan

Complexion : Medium

Nationality : Filipino

Religion : Islam

Height : 5’0”

Weight : 108 lbs.

Academic:

Elementary : Tangcal Central Elementary School

High School : MSU Binidayan Community High School

Skills:

Cooking

CURRICULUM VITAE



Personal:

Name : Hasnia M. Basher

Gender : Female

Civil Status : Single

Address : Bayanihan Village, Sta. Elena

Place of Birth : Campo Tubaran, Lanao del Norte

Date of Birth : January 3, 2008

Age : 18

Parents : Saida M. Tampugao
: Marining L. Basher

Complexion : Medium

Nationality : Filipino

Religion : Islam

Height : 4’7”

Weight : 110 lbs.

Academic:

Elementary : Bagong Silang Elementary School

High School : Tomas Cabili National High School

CURRICULUM VITAE



Personal:

Name : Norjana A. Ela

Gender : Female

Civil Status : Single

Address : Steel Town, Sta. Elena

Place of Birth : Pantao Raya, Poonapiagapo, Lanao del Norte

Date of Birth : July 28, 2008

Age : 18

Parents : Ela Mamasao
: Sobaida Ampa

Complexion : Medium

Nationality : Filipino

Religion : Islam

Height : 4'9"

Weight : 86 lbs.

Academic:

Elementary : Poonapiagapo Integrated School

High School : Tomas Cabili National High School

CURRICULUM VITAE



Personal:

Name : Asrima D. Osop

Gender : Female

Civil Status : Single

Address : Nangka Balo-i

Place of Birth : Balo-i, Lanao del Norte

Date of Birth : March 17, 2007

Age : 19

Parents : Carima D. Radiab
: Osop Hadji Solaiman

Complexion : Medium

Nationality : Filipino

Religion : Islam

Height : 5’1”

Weight : 103 lbs.

Academic:

Elementary : Nangka Elementary School

High School : Tomas Cabili National High School

CURRICULUM VITAE



Personal:

Name : Charlyn Jen Cristen A. Sanchez

Gender : Female

Civil Status : Single

Address : Purok 21 Tominobo, Scions

Place of Birth : Purok 21 Tominobo, Scions

Date of Birth : January 14, 2010

Age : 16

Parents : Jennifer A. Sanchez
: Arnel P. Sanchez

Complexion : Medium

Nationality : Filipino

Religion : Roman Catholic

Height : 4’10”

Weight : 88 lbs.

Academic:

Elementary : Tomas Cabili Central School

High School : Tomas Cabili National High School

Skills:

Drawing